



Sofia Krasovskaya

Curriculum Vitae

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[Cortexophie.github.io](https://cortexophie.github.io)

Education and training

PhD

University of Iceland

Reykjavik, Iceland

Field of study: Cognitive Neuroscience

[2021 – Oct 2023]

Thesis: Improvement of spatiotopic models of vision with retinotopic input and vector based saccadic generation

Supervisors: Dr. Árni Kristjánsson, Dr. W. Joseph MacInnes

PhD

HSE University

Moscow, Russia

Field of study: Cognitive Neuroscience
(Psychophysiology)

[2018 – 2022]

Terminated &
transferred to
University of
Iceland

Thesis: Improvement of spatiotopic models of vision with retinotopic input and vector based saccadic generation

Supervisors: Dr. Árni Kristjánsson, Dr. W. Joseph MacInnes

MSc

HSE University

Moscow, Russia

Field of study: Cognitive Neuroscience

[2016 – 2018]

Thesis: Deep learning neural nets as a model of saccadic generation

Supervisor: Dr. W. Joseph MacInnes

BA

Moscow State Linguistic University (MSLU)

Moscow, Russia

Field of study: Linguistics

[2009 – 2013]

Thesis: The correlation between prosodic and non-verbal means in crosscultural communication

Supervisor: Dr. Nadezhda Tsibulya

Publications

Krasovskaya, S., Coughlan J. M., Teng S (submitted). Modeling human echolocation using a Kalman filter.

Krasovskaya, S., Kristjánsson, Á., & MacInnes, W. J. (2024). Assessing the Size of the Functional Field of View in a Gaze-Contingent Search Paradigm. *ACM Hum.-Comput. Interact.*, 8, 17. <https://doi.org/10.1145/3655597>

Krasovskaya, S., Kristjánsson, Á., & MacInnes, W.J. (2023). Microsaccade rate activity during the preparation of pro- and antisaccades. *Attention, Perception, & Psychophysics*. <https://doi.org/10.3758/s13414-023-02731-3>

Krasovskaya, S., Zhulikov, G., & MacInnes, W.J. (2021). Deep Learning Neural Networks as a Component of a Model of Saccadic Generation. *PsyArXiv*. <https://doi.org/10.31234/osf.io/zr6ma>

Merzon, L., Malevich, T., Zhulikov, G., **Krasovskaya, S.,** & MacInnes, W. J. (2020). Temporal Limitations of the Standard Leaky Integrate and Fire Model. *Brain Sciences*, 10(1), Article 1. <https://doi.org/10.3390/brainsci10010016>

Krasovskaya, S., & MacInnes, W. J. (2019). Saliency Models: A Computational Cognitive Neuroscience Review. *Vision*, 3(4), Article 4. <https://doi.org/10.3390/vision3040056>

Conferences

23rd International Multisensory Research Forum (IMRF)

[Durham, UK, July 15 – 18, 2025]

A Kalman Filter model of echo-guided head movement.

The Cognitive Neuroscience Society (CNS) 32nd Annual Meeting

[Boston, Massachusetts, March 29 – April 1, 2025]

Modeling target search in blind echolocators using a Kalman Filter with realistic exploratory behavior simulations

ACM Symposium on Eye Tracking Research & Applications (ETRA)

[Glasgow, UK, June 4 – 7, 2024]

Presentation: Assessing the Size of the Functional Field of View in a Gaze-Contingent Search Paradigm

44th annual European Conference on Visual Perception (ECVP)

[Nijmegen, the Netherlands, Aug 27 – Sep 1, 2022]

Presentation (poster): The effect of task difficulty on the dynamics of functional field of view adaptability: a gaze-contingent study

43rd annual European Conference on Visual Perception (ECVP)

[Online, Aug 22 – 27, 2021]

Presentation (poster): The effect of task difficulty on the dynamics of functional field of view adaptability

Psychonomics Annual Meeting

[Online, Nov 19 – 22, 2020]

Presentation (poster): Microsaccade suppression as a measure of oculomotor inhibition in the antisaccade task

20th European Conference on Eye Movements (ECM)

[Alicante, Spain, Aug 18 – 22, 2019]

Presentation (poster): Microsaccade suppression during antisaccade generation in the Posner cueing paradigm

XVI European Congress of Psychology (ECP)

[Moscow, Russia, July 02 – 05, 2019]

Presentation (poster): The effect of the antisaccade task on microsaccade suppression in the Posner cueing paradigm

41st annual European Conference on Visual Perception (ECVP)

[Trieste, Italy, Aug 26 – 30, 2018]

Presentation (poster): Temporal distribution of saccades with deep learning salience maps

40th annual European Conference on Visual Perception (ECVP)

[Berlin, Germany, Aug 27 – 31, 2017]

Presentation (poster): Training restricted Boltzmann machines to generate human-like eye movements

Honours and Awards

F32 Kirschstein-NRSA Fellowship Award, NIH/NEI

NIH/NEI [2026 – 2028]

Individual Fellowship application titled Computational Modelling of Perceptual Information Accumulation in Blind Individuals

Student travel award, University of Iceland

University of Iceland [2022]

Academic mobility between HSE University and the University of Iceland, HSE University and University of Iceland

University of Iceland, HSE University [2019-2022]

Winner of the HSE Advanced doctoral programme

HSE University [2019]

The Advanced doctoral programme provides an increased scholarship for doctoral graduates awarded each month for the duration of their studies.

The Programme is similar to international post-graduate programmes, such as PhD programmes at U.S. universities, and not only includes an obligatory educational part, but also presumes that students spend their full working day at their faculty, department, or research centre. This means that they take an active part in events organized by the research department. This requirement aims to focus doctoral students on academic development and better adjust them to the academic environment.

Best poster award

TMLSS [2018]

The Transylvanian Machine Learning Summer School, currently known as the Eastern European Machine Learning Summer School (EEMLSS) took place in Cluj-Napoca, Transylvania, Romania, from July 16-22, 2018. I received the best poster award for the poster titled *Deep-learning neural networks as a model of saccadic generation*.

Link: <https://www.hse.ru/en/neuroling/vml/news/222000510.html>

Winner of the HSE University Olympiad for University Students and Graduates

HSE University [2016]

Winners of this competition receive a full tuition waiver for their master's studies. This allowed me to continue my studies in the field of cognitive neuroscience.

Research experience

Cognition, Action & Neural Dynamics Lab (SKERI) [2025 – current]	Postdoctoral fellow
Icelandic Vision Lab (IVL) [2021 – 2023]	PhD researcher
Vision Modelling Lab (VML) [2018 – 2022]	PhD researcher
Attention, Computational Models and Eye Movements Research and Study Group [2017 – 2018]	Researcher
Attention, Computational Models and Eye Movements Research and Study Group [2017 – 2018]	Research assistant

Teaching experience

Cognitive Psychology [01/2024 – 05/2024]	Adjunct lecturer. Taught a course for undergrads at the University of Akureyri, Iceland
Icelandic Vision Lab (IVL) seminars [2019 – 2023]	Occasional scientific presentations at weekly IVL meetings
Vision Modelling Lab (VML) seminars [2018 – 2022]	Presenting at bi-weekly VML seminars
Eyetracking Methods and Techniques [2019 – 2022]	Practice sessions for undergraduate students at HSE University
Perception and Cognition [09/2022 – 11/2022]	Teaching assistant for the practical part of the class during which students learned to use Psychopy Builder for experiment setup
Methods of Eyetracking in the context of reading [04/10/2021]	Guest lecture for undergraduate students (Sirius Science and Technology University) <u>Topic:</u> <i>Eyetracking: studying the visual system and other possible applications</i>
Introduction to Cognitive Psychology [11/2020 – 12/2020]	Held seminars and exams for undergraduate students at HSE University

Technical skills

Programming and experiment setup

Python // PsychoPy // Matlab // Psychtoolbox-3 // Google Collab // OpenSesame // Java //
SR Research Eyelink hardware and software

Data analysis

R // Python // SPSS Amos

Miscellaneous

Brainstorm (EEG) // SPM 12 for fMRI // Deep Learning Neural Nets // Zotero // Mendeley // LaTeX

Additional training

Practical MEEG <i>[Oct 28 – 31, 2025]</i>	Participant of the MNE-Python workshop organised by the Cutting EEG association in Aix-en-Provence, France
Kriegeskorte Lab <i>[Feb 25, 2022]</i>	Invited talk: Online presentation of a part of my doctoral project
Academic mobility between HSE University and the University of Iceland <i>[2019 – 2022]</i>	Internships, Dual-PhD programme
Applied Cognitive Science Autumn School <i>[10/2020 – 11/2020]</i>	Participant
Oxford Autumn School in Neuroscience <i>[Sep 28 – 29, 2020]</i>	Participant
Neuromatch Academy <i>[Jul 10 – 29, 2020]</i>	Active participant of the Computational Neuroscience course
The VML Advanced Eye Tracking Workshop <i>[Oct 25 – 26, 2018]</i>	Part of the workshop organising team Presentation: Models of Fixation https://www.hse.ru/en/neuroling/vml/news/228855476.html
Transylvanian Machine Learning Summer School (TMLSS) <i>[Jul 16 – 22, 2018]</i>	Presentation: Deep-learning neural networks as a model of saccadic generation

Language skills

English	<i>Fluent</i>	Icelandic	<i>A1</i>
Russian	<i>Fluent</i>	Malay	<i>A1</i>
French	<i>B2</i>	Spanish	<i>A2</i>
Japanese	<i>Beginner</i>		

Interests and Hobbies

Sports	I am a dedicated athlete and have participated in official state swimming and squash competitions in Malaysia where I grew up. I am currently a competing powerlifter, a non-competing swimmer, and an amateur cold open water swimmer. I have a brown belt in Taekwondo, a green belt in Karate, and a Practitioner level 5 in Krav Maga. I enjoy an active lifestyle in general.
Travel	I love adventure and enjoy travelling and learning about the customs, languages and history of other cultures. Hiking, scuba diving, eating and spending time in nature make me happy.
Other interests	I like to learn, game, knit, read, explore, draw, write, drive, ride, cycle, cook, skate, listen to music, and so much more. I believe that balance in life is crucial for mental and physical well-being.